GOURAV CHAKRABORTY

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Education:

Ph.D., Computational Biophysics, IIT (ISM) Dhanbad, **Supervisor:** Dr. Niladri Patra ***2021-2026 (Expected)***

M.Sc. (First Class Honor), Chemistry, University of North Bengal ***2018-2020***

- *Specialization: Inorganic Chemistry*

B.Sc. (First Class Honor), Chemistry, Acharya Prafulla Chandra Roy Government College ***2015-2018***

- *Minor in Physics, Mathematics, English, Environmental Studies.*

PhD Projects:

- **Machine Learning-Guided Enhanced Sampling:** Developed a data-driven CV generation method coupled with hybrid OPES-eABF scheme for free-energy estimation of rare biomolecular events.
- **VAE-Driven Adaptive Sampling:** Implemented a VAE (variational autoencoders)-driven adaptive sampling method combined with Markov State Models (MSMs) to characterize the conformational and kinetic landscapes of pathological protein assemblies.
- **Dual-Inhibition Strategy for Pathological Proteins:** Designed a novel strategy to mitigate the co-aggregation of proteins (e.g.,α-synuclein, 14-3-3ζ) through quantitative characterization using Absolute Binding Free Energy, Random Accelerated MD (RAMD), and Umbrella Sampling.
- **ML-Optimized Oil Recovery Systems:** Integrated Molecular Dynamics (MD) with supervised Machine Learning (ML) models to optimize surfactant-nanoparticle interfaces for enhanced oil recovery applications.

Peer-Reviewed Publications:

1. **Chakraborty, G.**; Patra, N. A Hybrid OPES-eABF Framework for Efficient Exploration and Data-Driven Collective Variable Discovery in Complex Free-Energy Landscapes. *J. Chem. Inf. Model.* **2026**, *66* (1), 554–566. https://doi.org/10.1021/acs.jcim.5c02158.
2. **Chakraborty, G.**; Patra, N. Mapping Conformational and Kinetic Landscapes of 14-3-3σ/α-Synuclein Assembly via Variational Autoencoders and Markov State Models. *J. Chem. Inf. Model.* **2025**, *65* (23), 12846–12860. https://doi.org/10.1021/acs.jcim.5c02323
3. **Chakraborty, G.**; Chaudhary, A.; Patra, N. Dual Allosteric and Orthosteric Inhibition of 14-3-3ζ–A-synuclein Interaction: A Multiscale Simulation and Machine Learning Approach. Adv. Theory Simul. 2025, No. e01455. https://doi.org/10.1002/adts.202501455.
4. **Chakraborty, G.**; Deka, U.; Patra, N. Mechanistic Insights into Ligand-Mediated Modulation of 14-3-3η/Parkin Interaction through Molecular Dynamics and Enhanced Sampling Simulations. J. Phys. Chem. B 2025, 129 (35), 8902–8918. https://doi.org/10.1021/acs.jpcb.5c02124.
5. **Chakraborty, G.**; Patra, N. Elucidating the Molecular Basis of 14-3-3 Interaction with α-Synuclein: Insights from Molecular Dynamics Simulations and the Design of a Novel Protein-Protein Interaction Inhibitor. J. Phys. Chem. B 2024, 128 (29), 7068–7085. https://doi.org/10.1021/acs.jpcb.4c01743.

6. **Chakraborty, G.**; Ojha, K.; Mandal, A.; Patra, N. Optimizing Oil Detachment from Silica Surfaces Using Gemini Surfactants and Functionalized Silica Nanoparticles: A Combined Molecular Dynamics and Machine Learning Approach. Phys. Chem. Chem. Phys. 2025, 27 (8), 4429–4445. https://doi.org/10.1039/d4cp04724a.
7. **Chakraborty, G.**; Kolpe, M. S.; Nath, I. V. A.; Tiwari, A.; Jayaswal, P.; Patra, N. Computational Structure-Guided Approach to Simulate Delamanid and Pretomanid Binding to Mycobacterial F420 Redox Cycling Proteins: Identification of Key Determinants of Resistance. J. Biomol. Struct. Dyn. 2025, 43 (18), 11177–11197. https://doi.org/10.1080/07391102.2025.2498621.
8. **Chakraborty, G.**; Nath I, A., V.; Sharma, M.; Sheth, J.; Kori, M.; Tiwari, A.; Patra, N. In Silico Structural and Mechanical Insights into Bedaquiline Resistance Associated with High-Grade Non-Synonymous Mutations in atpE, mmpR5, and pepQ. J. Biomol. Struct. Dyn. 2024, 42 (20), 10937–10949. https://doi.org/10.1080/07391102.2023.2259486.
9. Basu, D.; Chakraborty, S.; **Chakraborty, G.**; Bhattacharya, S.; Patra, N.; Nayek, H. P. Solvent-Free Hydrosilylation of Carbonyl Compounds Catalyzed by Organotin(IV) Compounds. Dalton Trans. 2025, 54 (27), 10614–10626. https://doi.org/10.1039/d5dt01033k.
10. Negi, K.; Kumar, A.; **Chakraborty, G.**; Sahoo, S.; Patra, S.; Patra, N.; Bhutia, S. K.; Sahu, S. K. Rationally Designed Porous Self-Assembled Nanoparticles for Combinational Chemo-Photodynamic Therapy. Dalton Trans. 2024, 53 (43), 17465–17479. https://doi.org/10.1039/d4dt02454k.

Other Publications:

1. Dhankhar, K.; **Chakraborty, G.**; Bera, A.; Chowdhary, A.; Bandyopadhyay, A.; Mishra, N. C.; Patra, N.; Hazra, S. Structural Insights into the Mechanism of C3 Side Chain Fragmentation in Cephalosporins with SME-1 Class A Carbapenemase. *bioRxiv*, 2025. https://doi.org/10.1101/2025.07.17.665329.

Articles Under Preparation:

1. **Chakraborty, G**; Patra, N. Divergent Interconversion Pathways of Monomeric α-Synuclein: Kinetic Insights into Early-Stage Amyloidogenic Conversion across Pathological Isoforms. *Submitted to JCIM.*
2. **Chakraborty, G.**; Patra, N. Capturing Protein Structural Transitions Using Generative AI.

Technical Skills:

- **Softwares:** *GROMACS, NAMD, AMBER, PLUMED, COLVARS, MDAnalysis, MDTraj, TeraChem, Gaussian, VMD, Blender, Microsoft Office.*
- **Programming:** *Python, Bash, Tcl, C++.*
- **Molecular Modelling Suites:** *Materials Studio, Discovery Studio.*
- **Miscellaneous:** *Proficient in Linux-based operating systems, Experience with HPC environments and job schedulers (OpenPBS, Slurm)*

Teaching Experience:

- **Teaching Assistant**, MSc Quantum Chemistry. ***2024-2025***
- **Teaching Assistant**, MSc Practical Laboratory. ***2023-2024***
- **Teaching Assistant**, BTech Practical Laboratory ***2022-2023***
- **Subject Matter Expert**, Chemistry, Chegg India Private Limited ***2020-2022***

Awards and Achievements:

- **Best Poster Award**, Recent Trends in Chemical Science and Technology (RTCST), IIT Patna (2024).
- **Best Oral Presentation**, Green Steel Production via Decarbonization for Sustainable and Circular Economy (GSP), IIT (ISM) Dhanbad (2024).

- **Silver Medalist**, M.Sc. Chemistry, University of North Bengal (2020).
- **Qualified GATE** (Chemistry) – All India Rank 274 (2021), and 2258 (2020).
- **Mentor,** RSC Yusuf Hamied Chemistry Camp, IIT (ISM) Dhanbad (2024).
- **Volunteer,** Recent Trends in Chemical Sciences (RTCS), IIT (ISM) Dhanbad (2022).

References:

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- Prof. Saugata Hazra
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  Department of Biosciences and Bioengineering
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